Sarah Ali

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WHO AM I?

Electrical Engineering graduate from NUST with hands-on experience in embedded firmware development, UAV/drone prototyping, and hardware-software integration. Skilled in **ESP32/STM32 microcontrollers,** sensor interfacing, **PID/control systems**, basic **PCB prototyping**, and embedded system debugging. Proficient in **CAD design** using **Fusion 360** along with 3D printing for rapid prototyping. Experienced in building and testing UAV systems and embedded hardware solutions.

RELEVANT PROJECTS

- **Bicopter Flight Controller (ESP32, PID Control)**
  Designed, tuned, and executed a bi-copter flight controller using ESP32. Achieved stable flight, disturbance rejection, and precise PID gains for real-time control.
- **Self-Balancing Robot (ESP32, PID)**
  Engineered a near-complete PID-controlled balancing robot using ESP32—demonstrating real-time feedback integration and motion control.
- **Drone Auxiliary Control via ESP32 + GSM**
  In my FYP, I integrated an ESP32 GSM module as a secondary controller on Drone, enabling GSM-based OTP communication with users—enhancing drone interaction and control.

EXPERIENCE

Intern – DroNexas SEECS **(June – September 2025)**

- Designed 3 UAVs in Fusion 360 and 3D printed prototypes.
- Contributed to the **Tethered Drone System** showcased on 14th August Parade.
- Gained hands-on experience in **flight controllers, ESCs, and 3D design**.

Quality Assurance Engineer – 9t5 Pvt. Ltd., Lahore **(June – September 2024)**

- Designed and executed test plans and cases for web and mobile apps.
- Performed **manual and automated testing** (functional, regression, performance).
- Collaborated with developers to resolve bugs, improving release quality.

EDUCATION

Degree	Year	Institute	Grades
BS. Electrical Engineering	2026	NUST, Islamabad	CGPA: 3.3

EMBEDDED SYSTEM PROJECTS

PID Tuned Flight Controller of Bicopter –ESP32	**Bluetooth Controlled Circuit Board -Arduino**
Drone Auxiliary Control-ESP32 + GSM	**Temp Controlled Cooling System**
Tethered Drone System (Quadcopter)	**Voltage Overcharge Protection**
Arduino based Bluetooth Control Car	**Wireless Charger – using PCB**
Digital Voting Machine- FPGA	**Digital Multimeter- Arduino**

TECHNICAL SKILLS

- **Design & CAD:** Fusion 360, SolidWorks, AutoCad, 3D printing workflows.
- **Microcontrollers**: Extensive experience with Arduino **ESP32**, familiar with STM32,FPGA architectures
- **Control Systems**: PID tuning, real-time loop control, sensor integration
- **Development Tools**: Familiar with **VS Code**, **Arduino IDE, STM32Cube, Quartus etc.**
- **Hardware Design**: PCB prototyping, sensor interfacing, power optimization
- **Embedded Software**: C/C++ programming, real-time debugging, Git version control
- **Simulation & Analysis**: MATLAB/Simulink, ANSYS MoterCad (basic)